Holly Pasquinelli
Lab 4

Windows Forensics Part I

Non-technical Summary

In Part 1 of this lab, we explored windows commands that help us retrieve information from systems such as user information, privileges, and groups. We can also use many of the commands to explore system information, specifically CPUs, architecture, details of operating systems, name, version, manufacturer, system type, boot devices, total physical memory, page file location (virtual memory), network cards, and IP addresses. For the purpose of the commands in part 1, we explored the collection of volatile information. This means that data is lost when the computer is turned off. In digital forensics, this is important because these commands can help us retrieve the data in real time, and store it before the system is turned off. With this data we are able to do things like build a timeline, retrieve a list of processes, networks, users, and a memory dump to assist in putting an investigation together.

Technical Report

In class, we first looked at a few different commands that retrieve information. Using the whoami command we can display the user (/user), groups (/groups), privileges, and SID. This helps us gain more information about the user for security. A hacker may want to access a group with more priviledges.

Using the PsExec64.exe command, we are able to execute processes on a different system. This is helpful when it comes to forensics because we are able to execute commands on a system that may have been compromised and retrieve information without being on the physical system and modifying evidence.

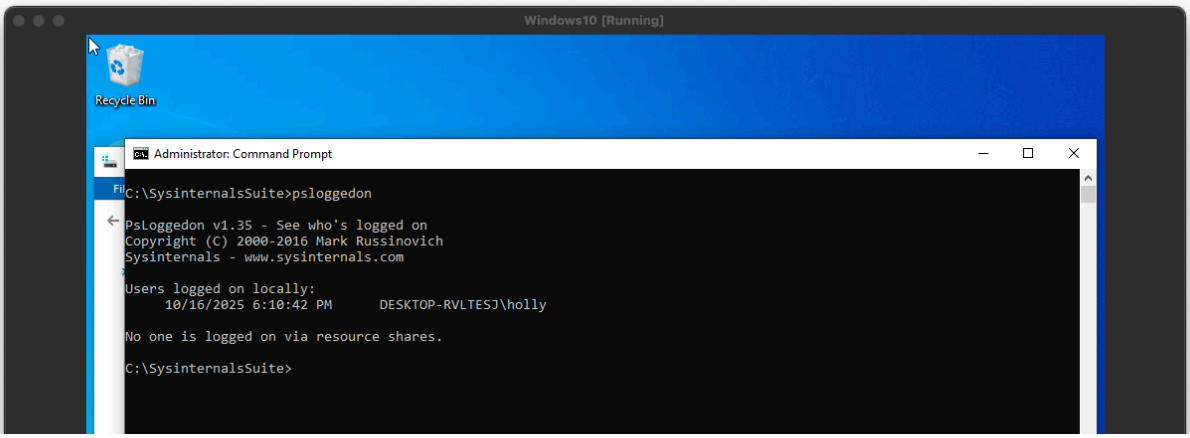
Another very useful command is Systeminfo, which displays information about CPUs, architecture, details of the operating system, version, manufacturer, system type, boot devices, total physical memory, page file location (virtual memory), network cards, and IP addresses.

Diskpart allows us to retrieve disk information such as partitions, volumes, and disks. Here, we are able to create a volume, partition, or virtual disk. A partition is the logical splitting of a hard disk. Listdisk gives us additional information such as disks, size, and status of physical disks. Listvolume is similar, with respect to volume, and we can determine if a user can access a volume by looking at the ltr column. If there is no ltr listed, the user does not have access.

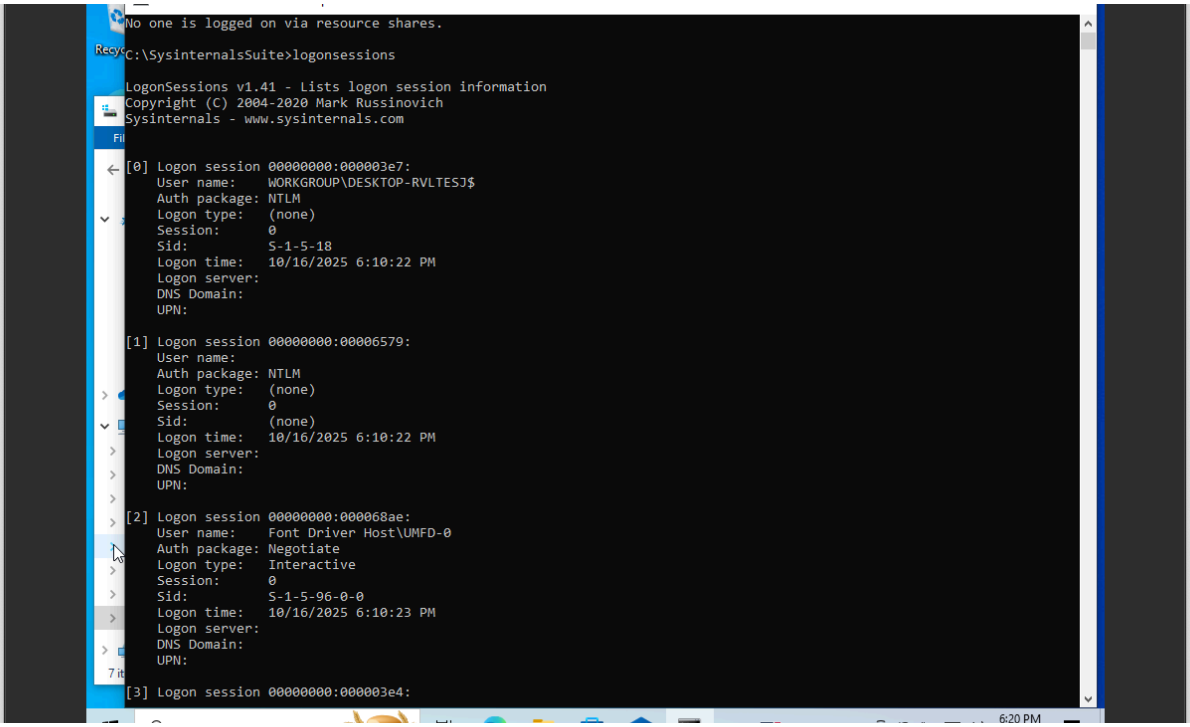
We also learned that windows stores values in the registry, using environment variables such as system, user, process. The registry is where we can access every connection, hardware, and in short, everything! The command regedit allows us to edit the registry. The command set is to create variables and setx for the machine and user.

Hackers will sometimes store values in environment variables such as TMP, TEMP,etc, so these are important means to look for any signatures hackers may leave behind.

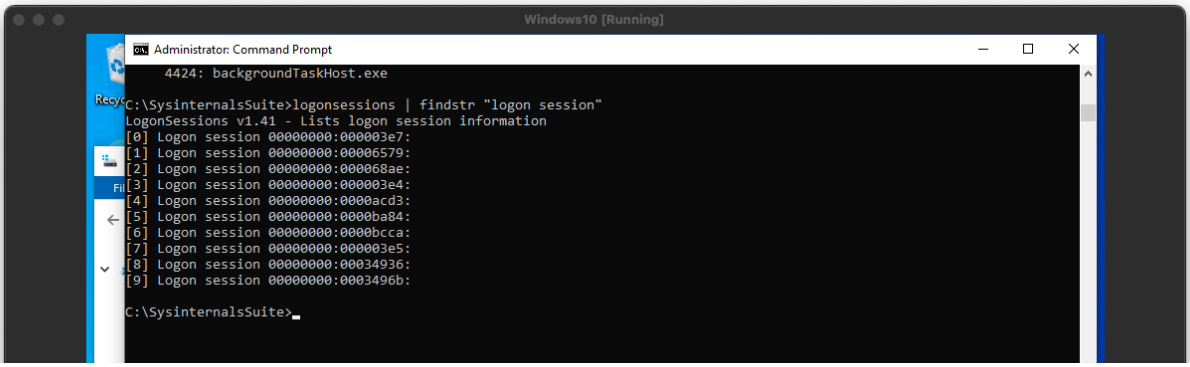
Below, we can see the psloggedon command demonstrated. These commands use the SysinternalsSuite. Here, we are able to see all users logged onto the system, including the network. We see here that no one is logged on via resource shares, so no one is logged onto the network remotely.



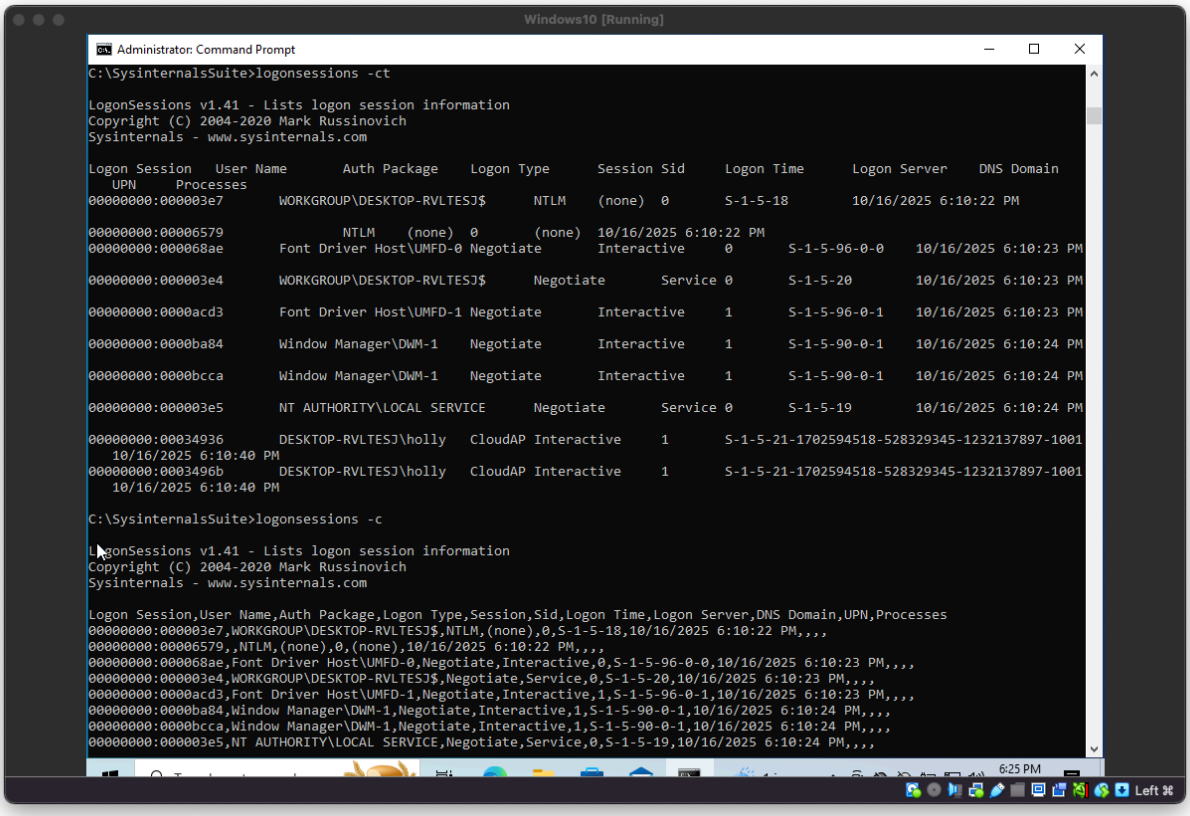
Using logonsessions, we are able to see the logon session information. While it is common to think that only one session should appear, the system takes into account other sessions such as admin, as well as background processes. We can use -p to list all processes running in each session, along with session information.



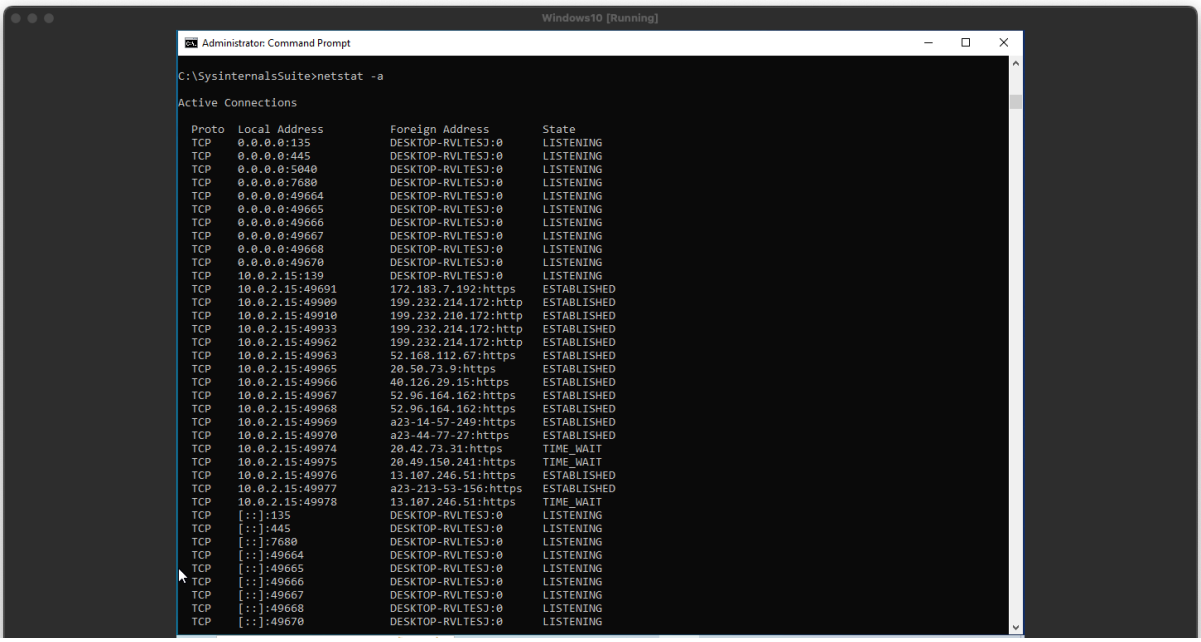
We can also truncate the output of logonsessions to specify only the lines containing “logon session” using a pipe and the findstr command. This allows us to see each session more clearly without the rest of the session and process information.



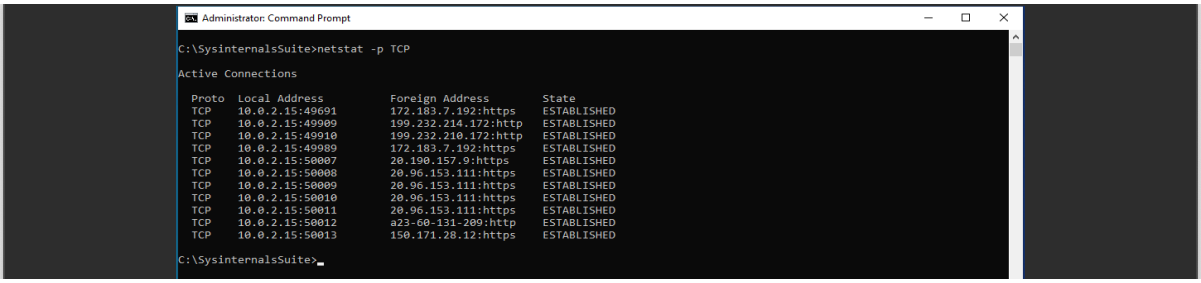
-ct will display all the information as tab delimited values, and -c will display the information as comma separated values as shown below.



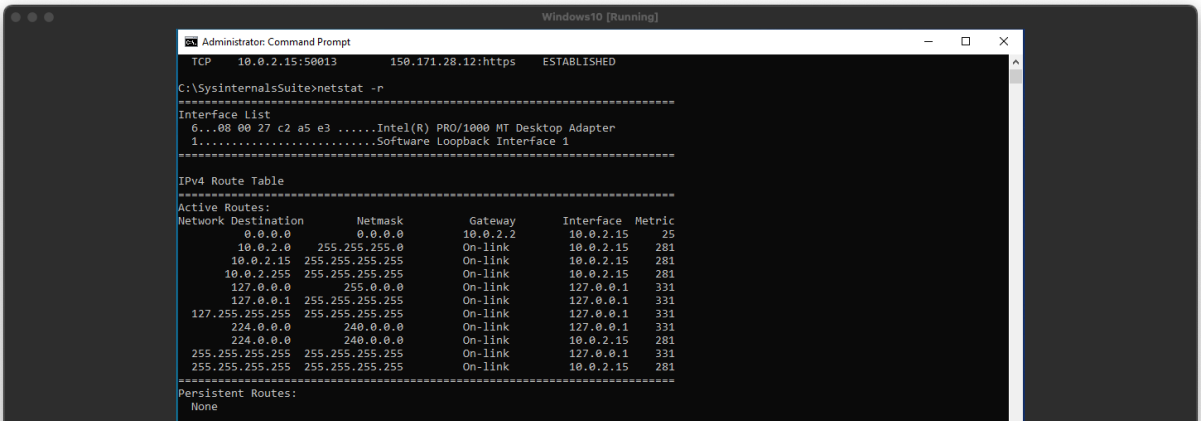
The netstat command collects all information about network connections. Using netstat -a displays all information such as the protocol, local/foreign IP address, port, and network state.



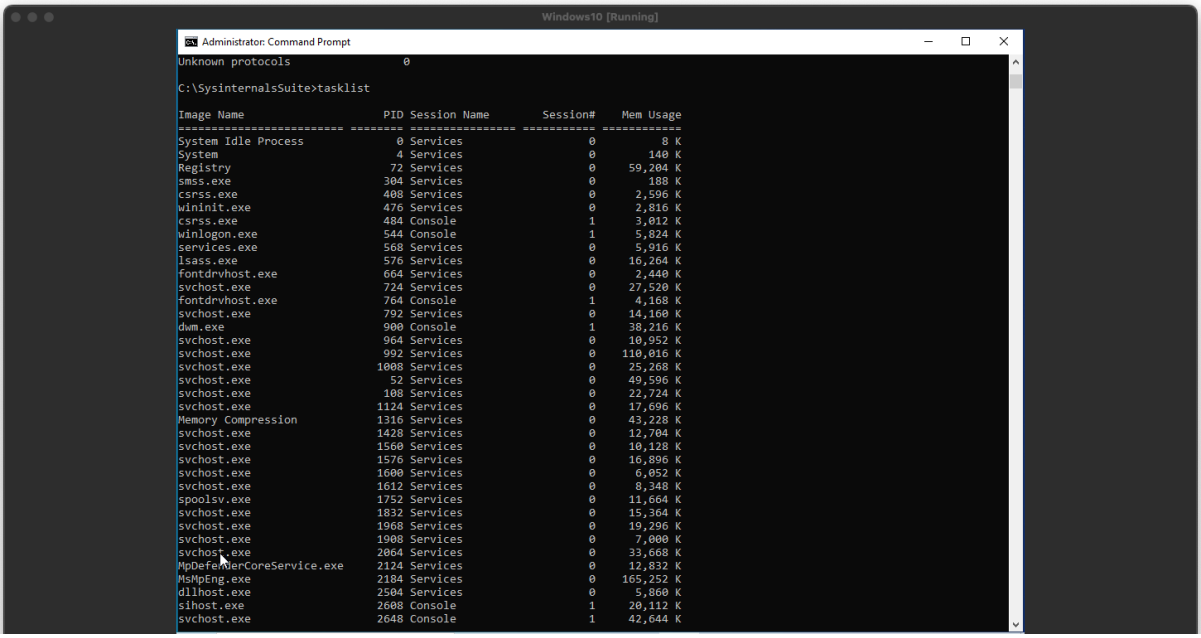
We can filter the protocol by using -p [protocol]. Here we see the TCP protocol.



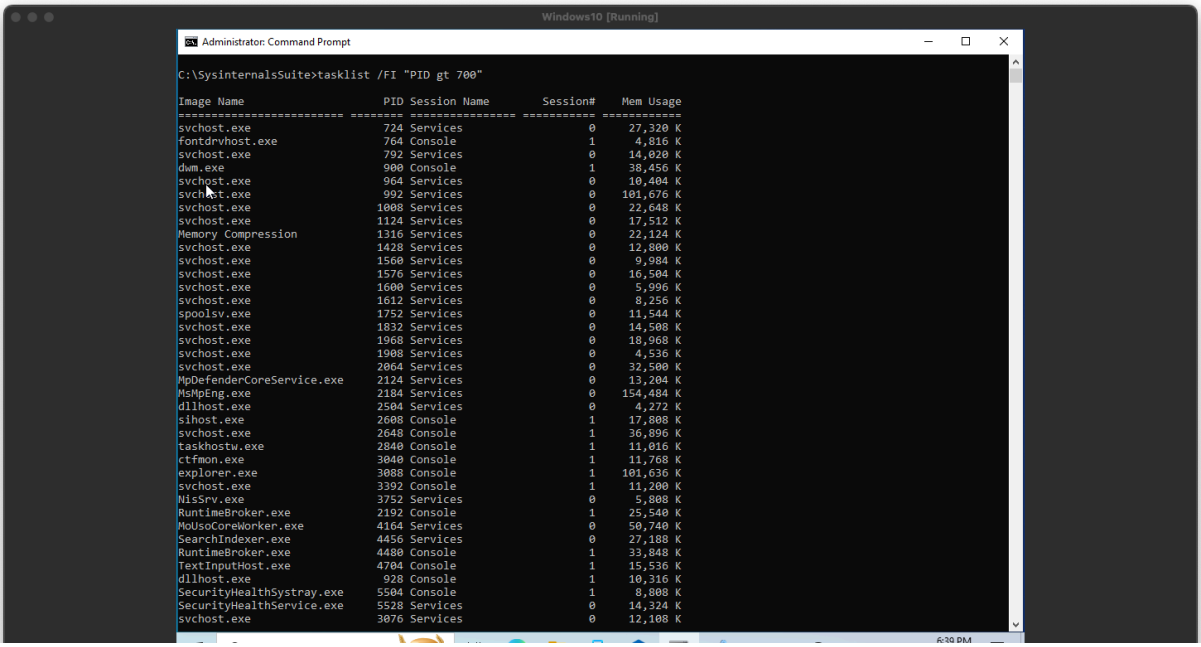
We can display the routing table using -r, and -e to display network statistics.



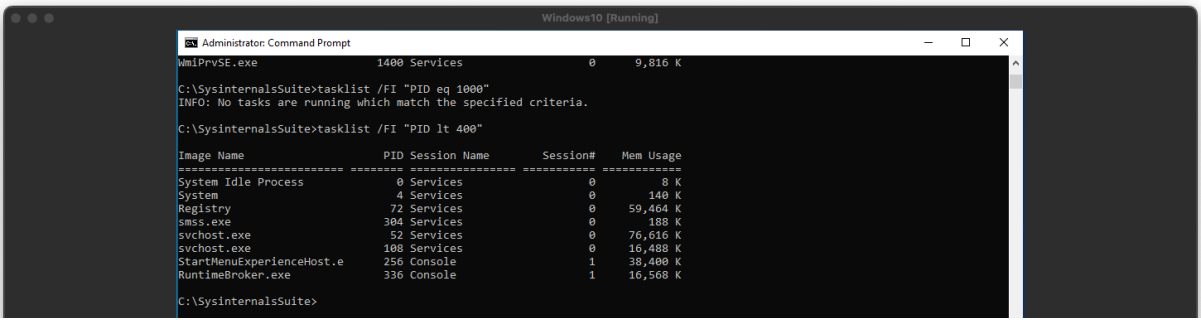
The tasklist command collects information about and displays running processes on a system. Here we can see the image name, PID, session name, session #, and memory usage.



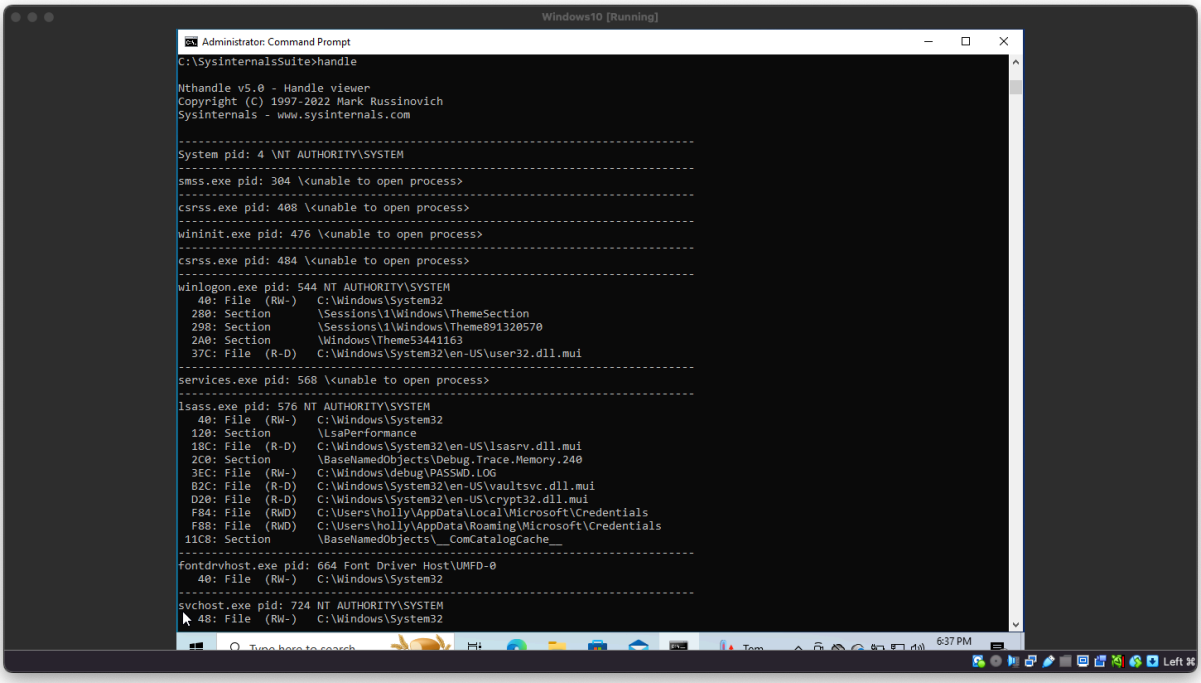
We can filter through these processes by using the /FI, or filter specifier. Below, we can see /FI “PID gt 700”, which filters processes with PID greater than 700.



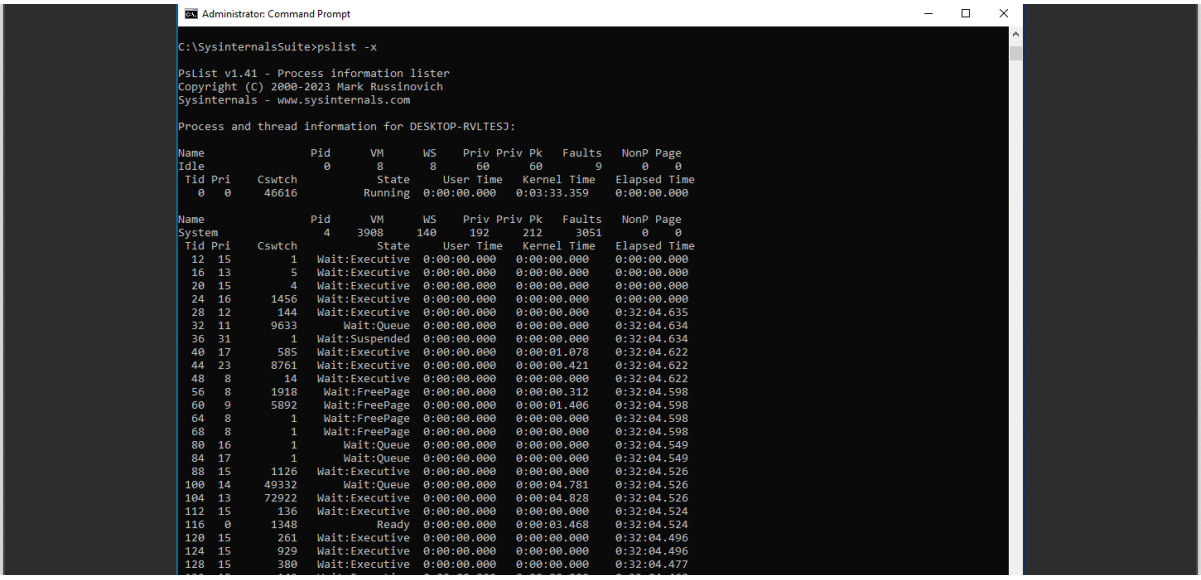
We can also filter PID values eq, or equal to. As we can see below there are no PID values that match 1000. We can also filter by using lt, or less than, shown below as well.



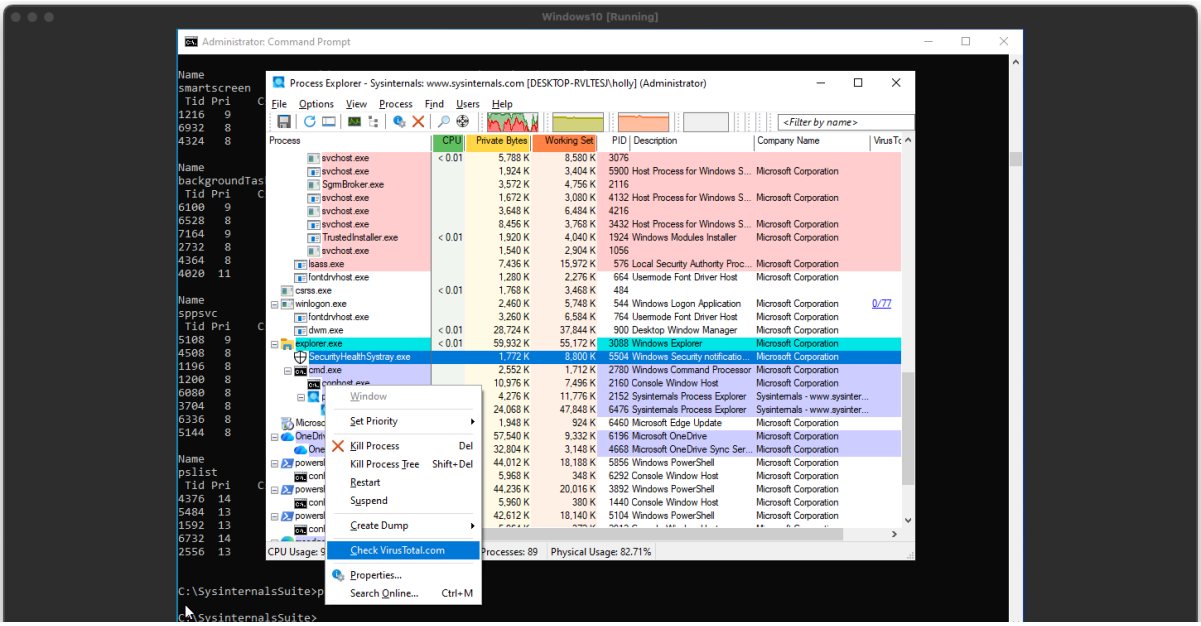
The handle viewer can be accessed using the handle command. Handle viewer allows us to see what directories or files processes have open. Without any filters, as we see below, it will list all handles with open files in the system.



Using pslist -x we can collect information about processes. Below, we use -x to display memory, threads, and PID. We can use -d for thread, and -m for memory separately.



The last part of this lab looked into procexp. Procexp allows us to collect information and manage all processes running on a system. We can see at the top it lists CPU,process, and memory usage. This allows us to create a memory dump, and check it for viruses.



References

1. Lecture Notes Windows Forensics - Part 1
2. Lab 6 Part 1 pdf
3. Windows tasklist
4. Windows handle